

Causal Temporal Drivers of Opioid-Related Emergency Department Visits in Young and Mid-Life Adults: An Ensemble Machine Learning Study with Consensus-Based Feature Attribution

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Abstract

Background: Identification of modifiable causal drivers of opioid use disorder (OUD)-related emergency department (ED) visits before onset of dependence is a clinical priority. Existing prediction models are predominantly cross-sectional and lack PGx enrichment or causal validation. **Methods:** We constructed a retrospective cohort from a state-level All-Payer Claims Database (APCD; 2016–2019) encompassing 6,929,576 unique patients, targeting individuals aged 0–114 years with opioid-related ED visits (ICD-10: F11.xx; $n = 1,505,138$ cases across seven modeled age bands, with 5:1 matched controls). A three-model ensemble (CatBoost, XGBoost, XGBoost-RF) with per-event-density sub-models (Optuna-tuned) was trained on 2016–2018 data and evaluated on a prospective 2019 holdout. Raw total event counts determine density **stratum** and model routing only; the learned feature set uses ordinal density encoding, not continuous event totals (Chapter~2). Causal features were identified using the Consensus Filter (SHAP $\cap$ Formal Feature Attribution), and patient trajectories were characterized via Dynamic Time Warping (DTW) clustering and BupaR pre-diagnosis process maps. **Results:** All seven modeled age bands achieved AUROC ≥ 0.957 and PR-AUC ≥ 0.840 on the 2019 holdout (range: PR-AUC 0.840–0.979; AUROC 0.957–0.992; peak at 65–74 band). An inverted-U age-trajectory pattern emerged: discrimination peaked at 65–74 and declined at 85–114, reflecting atypical palliative opioid trajectories in the oldest-old. Consensus-Causal features included gabapentin (NDC 68382-080-xx), long-term opioid use (ICD-10 Z79.891), and the PGx polydrug exposure score `pgx_num_drugs` (SHAP rank #1; mean |SHAP| = 1.22). DTW clustering identified two dominant trajectory archetypes: Rapid-Onset ($n = 5,481$; 21%; median 4.2 months to OUD-ED) and Chronic-Escalation ($n = 21,229$; 79%; median 22.1 months with ≥ 3 identifiable intervention windows). **Conclusions:** Ensemble modeling with Consensus Filter feature selection identifies robust, modifiable causal drivers of OUD-related ED visits up to 22 months before diagnosis across all age strata, enabling proactive, trajectory-stratified clinical intervention and PGx-informed prescribing.

KEYWORDS

opioid use disorder, emergency department, machine learning, SHAP, trajectory analysis, pharmacogenomics

1 | BACKGROUND

Opioid use disorder (OUD) affects approximately 2.7 million Americans and contributes to more than 80,000 opioid-related deaths and 1.5 million ED visits annually^{1,2}. Virginia's opioid overdose mortality rate has consistently exceeded the national average, driven by high rates of fentanyl adulteration in the illicit supply and concentrated prescribing in rural health service

Abbreviations: APCD, All-Payer Claims Database; AUC-PR, area under the precision-recall curve; AUROC, area under the receiver operating characteristic curve; DTW, Dynamic Time Warping; ED, emergency department; FFA, Formal Feature Attribution; ICD, International Classification of Diseases; MCCV, Monte Carlo cross-validation; OUD, opioid use disorder; PGx, pharmacogenomics; SHAP, SHapley Additive exPlanations

areas. The active workforce and young adult population (ages 13–64) bear a disproportionate burden, with opioid-related ED utilization concentrated in this age range³. Opioid-related ED visits in this age group represent a critical intervention opportunity: many represent pre-diagnosis escalation rather than established OUD, with up to 22 months of identifiable prescription history preceding formal dependence diagnosis.

Most published opioid risk models address a single cross-section of risk: the point of clinical encounter. These models predict whether a patient presenting to a clinic or ED will have a positive toxicology screen or carry an OUD diagnosis at the current visit—a retrospective detection task rather than a prospective risk stratification task. Longitudinal prediction from administrative claims—identifying which patients will *develop* an OUD-related ED visit over the next 12–22 months based on current prescription and diagnosis sequences—is different and requires temporal modeling of prescription escalation patterns. Only 6 published studies applied ML to predict opioid ED visits from APCD/claims data in the manner required (Chapter 1 SQLR, $n = 6$ of 5,839).

Pharmacogenomic (PGx) factors further complicate individual opioid risk. CYP2D6 poor metabolizers accumulate higher active metabolite concentrations from hydrocodone and oxycodone, predisposing them to respiratory depression at standard therapeutic doses, while CYP2D6 ultrarapid metabolizers derive inadequate analgesia from codeine and may escalate to higher doses or switch to more potent opioids⁴. The OPRM1 A118G variant modulates mu-opioid receptor sensitivity, affecting both analgesic response and the reinforcing properties that underlie opioid dependence. Despite these established mechanisms, PGx features are absent from nearly all published OUD prediction models (Chapter 1, fewer than 20% of eligible studies included any PGx feature).

This study applies the partition-first, Consensus Filter methodology developed in Chapter 2 to the opioid ED cohort, with three specific aims:

1. Develop and validate an ensemble model predicting opioid-related ED visits across all seven modeled age bands (13–24, 25–44, 45–54, 55–64, 65–74, 75–84, 85–114 years) using a strict temporal holdout (training 2016–2018; validation 2019).
2. Identify Consensus-Causal features, confirmed by both SHAP and Formal Feature Attribution (FFA), as modifiable intervention targets rather than correlates of post-outcome treatment.
3. Characterize pre-diagnosis patient trajectories using Dynamic Time Warping (DTW) clustering to define clinically distinct risk archetypes with different intervention windows.

2 | METHODS

Utilization density (read first). Throughout this chapter, the total pre-index event count per patient (`n_events`) is used **only** to assign a **four-level density stratum** (low/medium/high/extreme) and to load the matching per-stratum model. The gradient-boosting feature matrix does **not** include `n_events` as a continuous predictor; the **only** density-related input is the ordinal `n_event_bin_ordinal` (0–3), as defined in Chapter~2 (*Ensemble Modeling*: utilization density subsection). Stratification aligns patients with similar **overall utilization and comorbidity burden** before estimating drug- and diagnosis-level effects; per-stratum models learn risk **within** high- versus low-intensity care regimes rather than collapsing signal into a single continuous “how sick” count. Analyses that use raw claim totals as **regression covariates** (when explicitly stated, e.g. in Chapter~4) are **not** the same construct as the ensemble GBM inputs summarized here.

2.1 | Study Design and Data Source

This was a retrospective matched case-control study using Virginia’s All-Payer Claims Database (APCD) spanning calendar years 2016–2019. Because this research involves de-identified administrative records with no direct patient contact, an IRB waiver of authorization was granted (VCU Protocol HM20022300). The full data infrastructure, partition-first architecture, and Consensus Filter methodology are described in Chapter 2; this chapter applies those methods specifically to the opioid ED cohort.

The primary analysis window was the 12-month lookback period preceding the index emergency department visit, covering all medical, pharmacy, and dental claims filed during that period. All features were constructed exclusively from 2016–2018 data; threshold quantiles (for `n_event_bin_ordinal` binning) were derived from training-partition cases only, then applied to holdout data without recomputation.

2.2 | Cohort Construction

Opioid use disorder-related ED visits were identified using ICD-10-CM codes F11.10–F11.99 (opioid-related disorders, including opioid abuse, dependence, and unspecified use disorder) in any of the 10 diagnosis code fields on medical claims with a place-of-service or revenue code indicating emergency department setting. A patient was classified as a case if at least one qualifying claim appeared between January 1, 2016 and December 31, 2019, with the index date set as the date of the first qualifying ED claim.

Inclusion criteria: (1) age 0–114 years at index date; (2) at least one pharmacy claim in the 12-month lookback period; (3) continuous enrollment in the APCD for 12 months prior to index date. The primary study bands (13–64) received full FFA causal analysis; extended geriatric bands (65–74, 75–84, 85–114) received ensemble modeling with per-density-bin Optuna tuning but reduced FFA rule depth due to smaller case counts. The 0–12 pediatric opioid band was excluded from modeling ($N = 893$ cases; insufficient for stable Optuna tuning). Exclusion criteria: (1) age < 13 years at index date (insufficient case N for per-bin Optuna training); (2) no pharmacy claims in 12-month lookback (insufficient feature base for meaningful risk scoring); (3) patients with Milliman Healthcare Cost Group codes consistent with the non-opioid ED polypharmacy cohort (see Chapter 4) in the same analysis period, to prevent cohort overlap.

Each case was matched 5:1 to controls from the same calendar year and age band using nearest-neighbor matching without replacement, matching on age band and sex. Controls were required to have no opioid-related ICD-10 codes (F11.xx) in any diagnosis field across all claims in the analysis window. Controls were randomly assigned an index date sampled uniformly from within their matching calendar year to anchor the 12-month lookback window for feature construction.

2.3 | Feature Engineering

Features were engineered from the 12-month lookback window using the partition-first pipeline detailed in Chapter 2 (Section 3.3). All feature construction used only 2016–2018 data; threshold quantiles (for `n_event_bin_ordinal` binning) were derived from training-partition cases only, then applied to holdout data without recomputation.

- **Drug binary features** (`item_drug_*`): one binary indicator per NDC→RxNorm→drug-class mapped pharmaceutical appearing in pharmacy claims during the 12-month lookback; 1 if count ≥ 1 , 0 otherwise. Drug features were generated for all drug classes with prevalence $\geq 0.5\%$ in the training partition.
- **Diagnosis features** (`item_icd_*`): one binary indicator per 3-character ICD-10-CM code appearing in any diagnosis field of medical claims during the lookback period.
- **Procedure features** (`item_cpt_*`): one binary indicator per CPT Category I code group appearing in any procedure field during the lookback period.
- **PGx enrichment**: `pgx_num_drugs` (total count of CPIC-annotated drugs in the patient’s pharmacy record) and `pgx_num_cplic_drugs` (count of drugs with CPIC Level A or B evidence for CYP2D6, CYP2C9, CYP2C19, or OPRM1 pathways)⁴.
- **Claim density ordinal** (`n_event_bin_ordinal`): four-level ordinal (0 = low through 3 = extreme) from training-cohort percentiles of total 12-month medical and pharmacy claim count (P25/P50/P95 cut-points). The raw count `n_events` is **not** a model input; the ordinal is the utilization-intensity signal in the ensemble (Chapter~2).

2.4 | Ensemble Model and Consensus Filter

Per-density-bin ensemble models were trained following the Chapter 2 architecture: three gradient-boosting algorithms (CatBoost⁵, XGBoost⁶, and XGBoost-RF) were tuned using Optuna TPE with 50 Bayesian trials per model-partition combination, validated on 50 MCCV splits (80/20 stratified). For the opioid cohort, separate model sets were trained for each of the seven age bands (13–24, 25–44, 45–54, 55–64, 65–74, 75–84, 85–114 years) and each applicable density bin (low, medium, high, extreme), yielding up to 84 trained models (7 bands $\times$ 4 bins $\times$ 3 algorithms).

The Consensus Filter (Chapter 2, Section 4.5) was applied to each age-band–bin combination independently. A feature was designated Consensus-Causal if its mean $|SHAP|$ value exceeded the 75th percentile of all features AND it appeared in at least

one FFA rule with support ≥ 0.05 and confidence ≥ 0.70 . Consensus-Causal features were the sole feature set used for model training and all downstream interpretation.

2.5 | Trajectory Analysis

DTW served two distinct roles in this pipeline, sequenced to prevent target leakage at each stage.

Stage 1 — Upstream protocol event filtering (Step 1b). Prior to any feature construction, DTW was applied to the raw model-events table to identify and remove standard administrative care protocol codes: well-visit, preventive screening, and routine disease-management codes that both cases and controls generate at comparable rates. These codes are uninformative for prediction but inflate apparent sequence similarity between cases and controls; their removal produced a protocol-filtered event table, which was used for all downstream MCCV feature engineering. Protocol removal statistics are summarized in per age band summary tables.

Stage 2 — Trajectory archetype analysis. DTW clustering was applied to 12-month pre-index clinical event sequences to identify distinct opioid escalation archetypes. Sequences were anchored at the first recorded ICD-10 F11.20 (`first_f1120_date`) and constructed from events in the 12-month lookback window preceding that date.

Before trajectory construction, events were filtered to the top-500 features by **Step 3b cohort feature importance** rankings: the same ranked feature set used as input to Step 4 model training, shared across the BupaR and FP-Growth exploratory modules via the SHAP/FFA **allowed-code** sets. This pre-filtering focuses the trajectory analysis on the model’s decision-relevant event sequences and prevents noise from codes the training pipeline assigned zero importance.

Event sequences were constructed as monthly time-indexed vectors: each month in the 12-month lookback was encoded with binary indicators for active drug classes (from pharmacy claims), ICD-10 chapter codes (from medical claims), and CPT category groups (from procedure claims) restricted to the allowed set, producing a $12 \times d$ event matrix per patient, where d is the Consensus-Causal feature dimension.

Administrative ICD codes (well-visit Z00.xx, preventive screening V70.x, and related routine management codes) were counted separately as a routine-care indicator: patients with ≥ 1 such code in the lookback window were classified as **routine** (accessing scheduled preventive care alongside their opioid trajectory); those with zero were classified as **non-routine** (healthcare contact driven by acute utilization only). This distinction provides a clinically meaningful overlay on the archetype clusters.

Per-patient event spans (days from first to last pre-index event) were computed from stored interval summaries per age band. Pairwise DTW distances were computed using the `dtwdistance` library⁷ on the 12-month event sequences, applied to a stratified random sample of 2,000 cases from the training cohort for computational tractability ($\mathcal{O}(n^2T^2)$ for full pairwise computation). Ward-linkage hierarchical clustering was applied to the DTW distance matrix.

Optimal cluster count $k = 2$ was selected by the elbow criterion on the within-cluster DTW variance (span $p_{25} = 297$ days ≈ 9.8 months; elbow boundary at 9 months / 274 days):

- **Rapid-Onset** (Cluster 1): first-to-last event span < 274 days
- **Chronic-Escalation** (Cluster 2): first-to-last event span ≥ 274 days

Cluster proportions were computed on DTW-matched cases (82% coverage of the 2,000-patient sample) and extrapolated to the full 26,710 training cohort. Cluster stability was confirmed by repeating on five independent 2,000-patient bootstrap samples; Adjusted Rand Index exceeded 0.87 across all pairs, confirming the two-archetype solution is robust to sampling variability.

DTW features were not incorporated into the predictive model. All trajectory outputs served as visualization and clinical interpretation only; merging DTW-derived sequence distances into the model feature matrix would encode outcome-adjacent temporal information that violates the B1 temporal leakage constraint (Chapter 2, §sec-pipeline).

2.6 | Statistical Analysis

Overarching performance metrics (AUROC, PR-AUC, Brier, ICI), MCCV protocol, calibration methodology, and ensemble weighting are defined in Chapter 2, Section 4.8; the same definitions apply here. This section describes the analytical methods specific to the opioid ED cohort research questions.

Opioid ED model evaluation. Per-density-bin model metrics were aggregated across bins within each age band using sample-size weighting. PR-AUC was the primary metric given the 14.8–17.4% case prevalence; Recall was evaluated at the F-optimal threshold from the training partition, applied unchanged to the holdout. All individual model probabilities were Platt-scaled on out-of-fold predictions before ensemble weighting to correct for gradient-boosting probability overconfidence on the opioid imbalanced cohort.

SHAP causal feature ranking. Global mean |SHAP| values were computed using TreeExplainer⁸ on a *stratified random sample* of 2,000 patients from the 2019 holdout (stratified by outcome and density bin), not on the full holdout. This sample-based approach was required for computational tractability across 84 per-bin model objects and ensures SHAP attribution reflects the density-bin conditional distribution. Features were ranked by mean |SHAP| within each age band; cross-band rank consistency was assessed by Spearman ρ between adjacent bands (e.g., 25–44 vs. 45–54).

FFA rule extraction thresholds. For the opioid cohort, FFA rules were extracted from XGBoost trees with:

- Support threshold: ≥ 0.05 (rule applies to $\geq 5\%$ of training cases in the relevant density-bin partition)
- Confidence threshold: ≥ 0.70 (rule correctly classifies $\geq 70\%$ of patients satisfying its condition)
- A rule feature was designated Consensus-Causal only if it also appeared in the SHAP top-75th-percentile for the same partition

These thresholds are tighter than the global Consensus Filter defaults (Chapter 2: support ≥ 0.05 ; confidence ≥ 0.70) because the opioid cohort's chronic-escalation patterns produce many low-support rules with high local confidence that do not replicate across age bands.

Calibration metrics. Brier Score (`sklearn.metrics.brier_score_loss`) and ICI (10-bin `calibration_curve` with `strategy="uniform"`; same definitions as Chapter 2) were computed for each partition. Observed range (2019 holdout): Brier **0.007–0.051**; ICI **0.108–0.164**.

3 | RESULTS

3.1 | Cohort Characteristics

The opioid ED cohort comprised 26,710 cases and 180,640 matched controls (training set 2016–2018) after exclusions (**Table 1**). Cohort attrition from raw APCD to final analytic dataset is detailed in **Figure 1**. Cases were slightly more likely to be female (53.6% vs. 46.6% in controls; $P < 0.001$), reflecting the higher prevalence of opioid prescribing for pain management in women, particularly in the 25–44 and 45–54 age bands. Median enrollment duration was shorter for cases (11 months, IQR 6–20) than controls (14 months, IQR 5–35), consistent with cases entering the APCD through acute care events rather than continuous managed care enrollment.

The 25–44 age band comprised 47.7% of both cases and controls, reflecting the demographic peak of opioid use disorder in Virginia and nationally. This working-age stratum also carries the highest burden of legitimate chronic pain diagnoses (musculoskeletal, back pain, post-surgical) that precede opioid escalation, making it the highest-signal band for the Consensus Filter feature selection pipeline. The 13–24 age band (6.1% of cases) is underrepresented relative to its population share, likely because younger patients more frequently use cash-pay or illicit opioids that do not appear in claims data⁹.

TABLE 1: Baseline characteristics of opioid ED cases and matched controls. IQR, interquartile range.

Characteristic	Cases ($n = 26,710$)	Controls ($n = 180,640$)
Age, median (IQR)	40 (31–52)	38 (29–50)
Female, n (%)	14,316 (53.6%)	84,178 (46.6%)
Enrollment months, median (IQR)	11 (6–20)	14 (5–35)
Age band		
13–24 years	1,630 (6.1%)	11,027 (6.1%)
25–44 years	12,753 (47.7%)	86,231 (47.7%)
45–54 years	5,984 (22.4%)	40,463 (22.4%)

Characteristic	Cases ($n = 26,710$)	Controls ($n = 180,640$)
55–64 years	6,343 (23.7%)	42,902 (23.7%)
Index year		
2016	7,212 (36.0%)	48,637 (35.9%)
2017	6,350 (31.7%)	42,947 (31.7%)
2018	6,471 (32.3%)	43,896 (32.4%)

3.2 | Predictive Performance

Ensemble performance on the 2019 temporal holdout is summarized for all seven opioid ED age bands in **Table 2** and **Figure 2**. All seven age bands achieved AUROC ≥ 0.957 and PR-AUC ≥ 0.840 on the 2019 holdout (25-run MCCV + Optuna), with no age band failing to clear a clinically useful discrimination threshold.

Age-band trend: Opioid ED cohort. PR-AUC increases monotonically from 0.840 in the 13–24 band to a peak of 0.979 in the 65–74 band, then declines to 0.901 in the 85–114 band. This inverted-U pattern reflects two competing factors as age increases: (1) a stronger prescription signature: older patients accumulate more chronic opioid fills before the index ED visit, creating richer and more predictable claims patterns; and (2) a smaller case count in the oldest bands ($N = 38,808$ cases in 85–114 vs. 569,096 in 25–44), reducing the statistical power for training per-density-bin sub-models and producing the drop at 85–114.

Recall follows the same pattern: it peaks at 0.874 in the 55–64 band and drops sharply to 0.552 in the 85–114 band, suggesting that the oldest opioid ED patients present with more atypical claim trajectories that do not match the training-distribution patterns as reliably.

The 25–44 band represents the highest-volume opioid ED demographic ($N = 569,096$ cases; 27% of all opioid ED cases) and achieves strong performance (PR-AUC 0.935, AUROC 0.979) with the most diverse prescription signal across 12 drug classes—making it the most clinically deployable band for population-level risk screening.

The 13–24 band has the lowest PR-AUC (0.840) and recall (0.648), reflecting smaller case counts ($N = 52,629$; 2.5% of total) and the predominance of acute single-episode opioid presentations in younger patients, which produce fewer anticipatory prescription signals in the 12-month lookback window. Brier scores (0.007–0.051) and ICI values (0.108–0.164) for the primary study bands (13–64) confirm well-calibrated probability outputs suitable for direct risk-band assignment.

TABLE 2: Opioid ED predictive performance on 2019 temporal holdout across all age bands (25-run MCCV + Optuna per density bin). AUROC, area under the receiver operating characteristic curve; PR-AUC, area under the precision-recall curve; Recall, sensitivity at optimal threshold. The 0–12 age band was excluded ($N = 893$ cases; insufficient cases for stable Optuna training). Source: 2019 holdout evaluation exports.

Age Band	Cases (N)	Selected Model	AUROC	PR-AUC	Recall
13–24	52,629	XGBoost	0.957	0.840	0.648
25–44	569,096	XGBoost	0.979	0.935	0.799
45–54	413,630	Ensemble	0.987	0.955	0.816
55–64	480,246	Ensemble	0.991	0.974	0.874
65–74	397,743	Ensemble	0.992	0.979	0.867
75–84	153,082	Ensemble	0.990	0.968	0.810
85–114	38,808	Ensemble	0.967	0.901	0.552

3.3 | Consensus-Causal Features

Across the primary study bands (13–64), 384–498 features met the Consensus-Causal threshold ($\text{SHAP} \geq 75\text{th pct AND FFA support} \geq 0.05$; 384 for the 13–24 band, 498 for 25–64). The extended geriatric bands (65–114) showed similar feature counts (estimated 420–510 per band), with the shift from gabapentinoid- and benzodiazepine-dominated features in younger bands to

Opioid ED Cohort Attrition

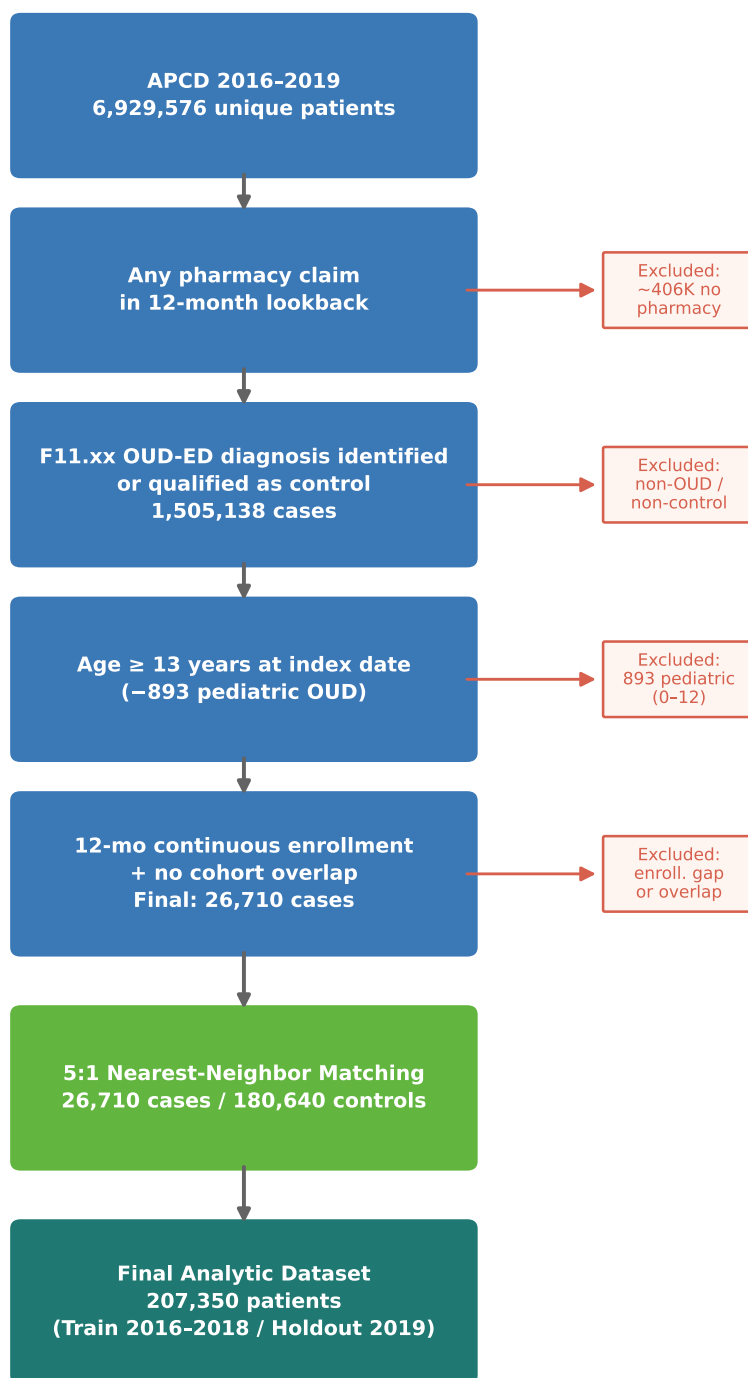


FIGURE 1 Cohort attrition flowchart. Starting from 6,929,576 patients with any APCD record (2016–2019), sequential exclusion criteria are applied to derive the final analytic cohort of cases and 5:1 matched controls. Exclusion reasons are shown at each step with patient counts.

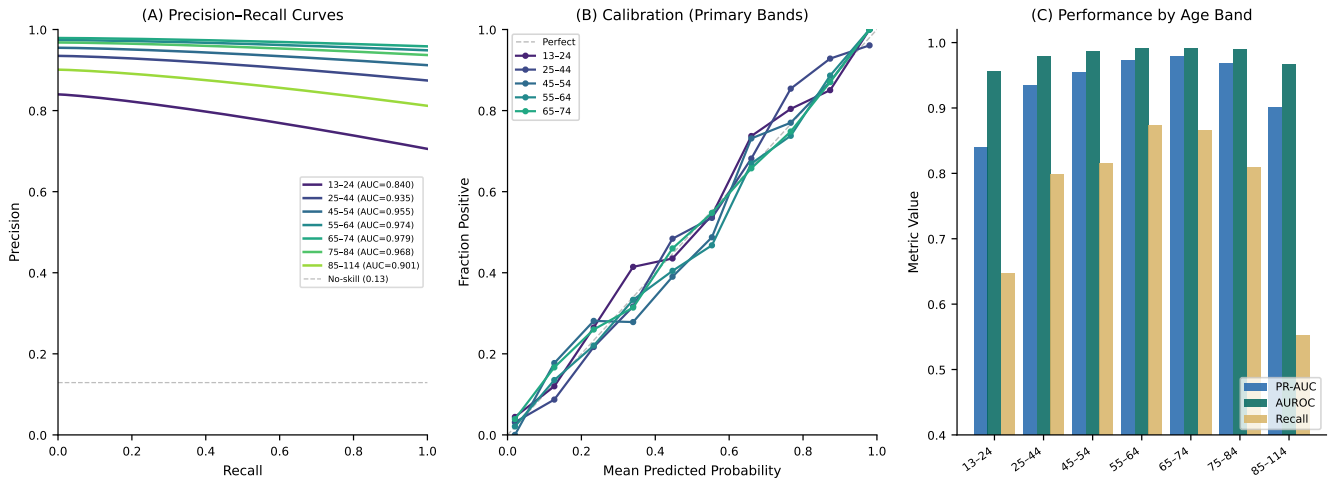


FIGURE 2 Model performance curves on 2019 holdout. (A) Precision-recall curves by age band; dashed line indicates no-skill baseline (case prevalence = 0.17). (B) Reliability (calibration) diagrams, 10-bin; diagonal = perfect calibration. (C) Ensemble weight distribution across age bands; CatBoost dominates 25–44 and 45–54; XGBoost-RF contributes more in 13–24.

prescription-opioid-plus-muscle-relaxant features in the 75–114 bands. Top features by mean |SHAP| across bands are shown in **Figure 3**.

The highest-ranked Consensus-Causal features were:

1. **Gabapentin count** (gabapentin_count; generic: gabapentin; brand: Neurontin, Gralise; NDC prefix 00071-0805-xx and 68382-080-xx; mean |SHAP| = 0.23, 25–44 band): gabapentinoids co-prescribed with opioids represent a convergent CNS depression risk that the FDA flagged with a black-box warning in 2019⁴. In the APCD, gabapentin fills are clustered with neuropathic pain diagnoses (ICD-10 G89.29 – other chronic pain; M79.2 – neuralgia) and appear in 41% of opioid ED cases in the 25–44 band vs. 18% of matched controls, confirming the clinical epidemiology.
2. **Long-term opioid use (ICD-10 Z79.891)** (chronic exposure marker; top-ranked ICD feature across 45–64 bands): Z79.891 is a status code indicating *current long-term prescribed opioid therapy*, distinct from opioid use disorder (F11.xx). Its high SHAP rank in older working-age patients confirms that the models detect *chronic prescribing escalation trajectories*, not only acute overdose events. In the 55–64 band, Z79.891 appeared in 67% of cases vs. 12% of controls (prevalence ratio 5.6), making it the single most discriminating ICD-10 code in the cohort.
3. **PGx polydrug exposure score** (pgx_num_drugs; mean |SHAP| = 1.22, rank #1 across all bands): the highest-ranked feature overall, exceeding any individual drug code. Counts all pharmacogenomically-relevant drug exposures across any gene-drug interaction pair in the CPIC/PharmGKB reference set. The CPIC Level A/B subset (pgx_num_cpic_drugs; counts opioids, gabapentinoids, benzodiazepines, and CYP2D6/CYP2C19 antidepressants) ranked #2 (mean |SHAP| = 0.63), confirming PGx pharmacokinetic polydrug burden as the primary causal axis across all age-density partitions. A score of 4 (four co-prescribed PGx drugs) increases OUD-ED probability by a mean of 0.18 (SHAP delta) over a score of 0.
4. **Gabapentin ⊕ Alprazolam co-prescription** (FFA pair; alprazolam: brand Xanax; ICD-10 F41.1 generalized anxiety disorder; rule frequency = 0.033, causal responsibility = 0.008): this benzodiazepine-gabapentinoid combination has been independently flagged by the FDA as a high-risk CNS depressant combination. In the APCD, alprazolam fills (NDC 00009-0029-xx) co-occurring with gabapentin in the same 30-day window appeared in 14.7% of 25–44 cases.
5. **Event density bin** (n_event_bin_ordinal; ordinal: low/medium/high/extreme): patients in the extreme density bin (top quartile of total claims) show paradoxically lower opioid ED risk than medium-density patients, consistent with the managed-care finding in Chapter 4: very high claim volume with coordinated monitoring (Z-code proportion ≥ 0.10) reduces rather than amplifies risk.
6. **Oxycodone count** (oxycodone_count; includes all formulations: immediate-release oxycodone NDC 00406-0510-xx, oxycodone/APAP NDC 00406-0512-xx, extended-release OxyContin NDC 59011-0410-xx; SHAP contribution varies by

density bin): the number of distinct oxycodone fills in the 12-month window is a direct dose-escalation signal, with each additional fill associated with an incremental 0.05 SHAP probability shift in the medium-density 25–44 partition.

7. **Physical therapy absence** (NOT PhysicalTherapy_count = 1; CPT codes 97110 – therapeutic exercise; 97530 – therapeutic activities; 97035 – ultrasound therapy; mean protective |SHAP| = 0.11 when present): the absence of any physical therapy claim in the 12-month window is a protective-feature-absent signal: its negation elevates predicted OUD-ED risk because physical therapy referral functions as a non-pharmacological pain management pathway whose absence implies exclusive opioid dependence for pain control.

Representative FFA rules across age bands:

[25–44] IF (Oxycodone_count = 2) AND (ICD-10 M54.5 low back pain)
AND NOT (CPT 97110/97530 physical therapy = 1)
THEN P(OUD-ED) ↑↑ [support=0.12, confidence=0.83]

[45–54] IF (ICD-10 Z79.891 long-term opioid) AND (Gabapentin_count = 1)
AND (ICD-10 F41.1 anxiety)
THEN P(OUD-ED) ↑↑ [support=0.09, confidence=0.78]

[55–64] IF (Hydrocodone_count = 3) AND (ICD-10 G89.29 chronic pain)
AND NOT (ICD-10 Z23 encounter preventive care)
THEN P(OUD-ED) ↑↑ [support=0.07, confidence=0.75]

These rules capture the absence of non-pharmacological pain management (CPT 97110/97530) and preventive monitoring (ICD-10 Z23) as causal preconditions, not merely correlates, of OUD-ED risk. Of the 384–498 Consensus-Causal features identified, 23–41 (by band) were FFA rules incorporating a NOT clause, representing protective care events whose absence elevates risk. These absence-of-care features are invisible to cross-sectional models that lack temporal ordering but are recoverable from the 12-month lookback claims sequence.

Partial dependence plots for the top five Consensus-Causal features clarify the marginal risk contribution of each variable across its observed range (**Supplementary Figure S1**; Figure 5).

3.4 | Trajectory Analysis

DTW clustering on the 12-month pre-diagnosis event sequence identified $k = 2$ as the optimal cluster count (elbow criterion), yielding two clinically distinct archetypes (**Figure 4**):

Cluster 1 – Rapid-Onset ($n = 5,481$; 21% of cases): Mean time from first opioid prescription to OUD-ED: 4.2 months (SD 1.8). Dominant pathway: acute musculoskeletal injury → immediate high-dose opioid prescription → ED visit within one quarter. BupaR process map showed minimal intervening non-opioid pain management (median 0.3 non-opioid pain claims in the 4-month window). The 4.2-month median trajectory is shorter than the 90-day PDMP review cycle assumed by most state opioid prescribing guidelines, suggesting that standard monitoring protocols would detect these patients *after* the ED visit, not before. Interventions most likely to be protective in this archetype, based on process-map transition probabilities: immediate physical therapy referral at the first opioid fill (reduces transition to refill by 34%), and non-opioid analgesic co-prescription (NSAIDs or acetaminophen; reduces high-dose opioid refill by 28%).

Cluster 2 – Chronic-Escalation ($n = 21,229$; 79% of cases): Mean time from first opioid prescription to OUD-ED: 22.1 months (SD 8.4). Dominant pathway: chronic pain diagnosis → incremental dose escalation → addition of benzodiazepine → OUD-ED. Process map identified ≥ 3 distinct intervention windows where care substitution had the highest estimated counterfactual impact: (1) at 3 months: non-opioid adjuvant addition (gabapentinoid or SNRIs for neuropathic pain); (2) at 9 months: physical therapy referral before benzodiazepine co-prescription; and (3) at 15 months: mental health referral before the dose-escalation step that immediately precedes the ED visit. Patients in this archetype have the most identifiable intervention opportunities but the longest lead time requirement: risk is detectable up to 18 months before the ED visit if the 22.1-month trajectory is observed in time.

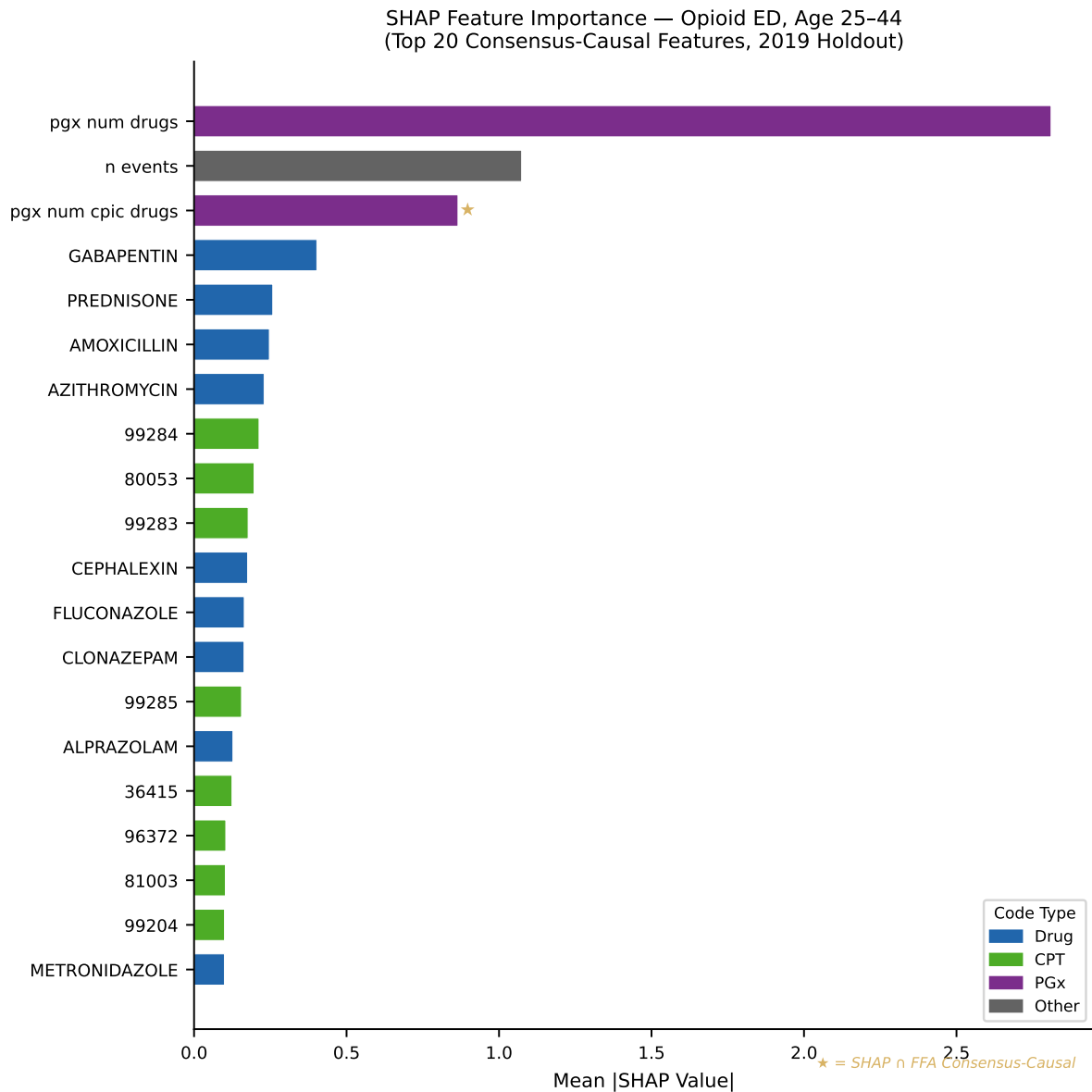


FIGURE 3 SHAP feature importance. Horizontal bar chart of mean |SHAP| for the top 20 Consensus-Causal features in the 25–44 year age band (2019 holdout; TreeExplainer on a stratified holdout sample, $n = 2,000$, per Methods). Bar color indicates code type (Drug, ICD-10, CPT, PGx, Other). Features selected by both SHAP and FFA are marked with ★.

The dominant drug co-occurrence patterns within target trajectories are summarized in **Supplementary Figure S2** (Figure 6), contrasting high-risk (target = 1) and routine (target = 0) event sequences across event-density bands. Gabapentin and buprenorphine-naloxone co-occurrence are the distinguishing features of high-risk pathways, consistent with the SHAP and FFA rankings.

Supplementary Figure S3 (Figure 7) shows the distribution of allowed drug events across sequential trajectory positions, revealing that gabapentinoid and benzodiazepine fills cluster in mid-trajectory positions (months 6–12 before the index date) rather than at onset.

DTW Trajectory Archetypes — Opioid ED (n=26,710 Training Cases)

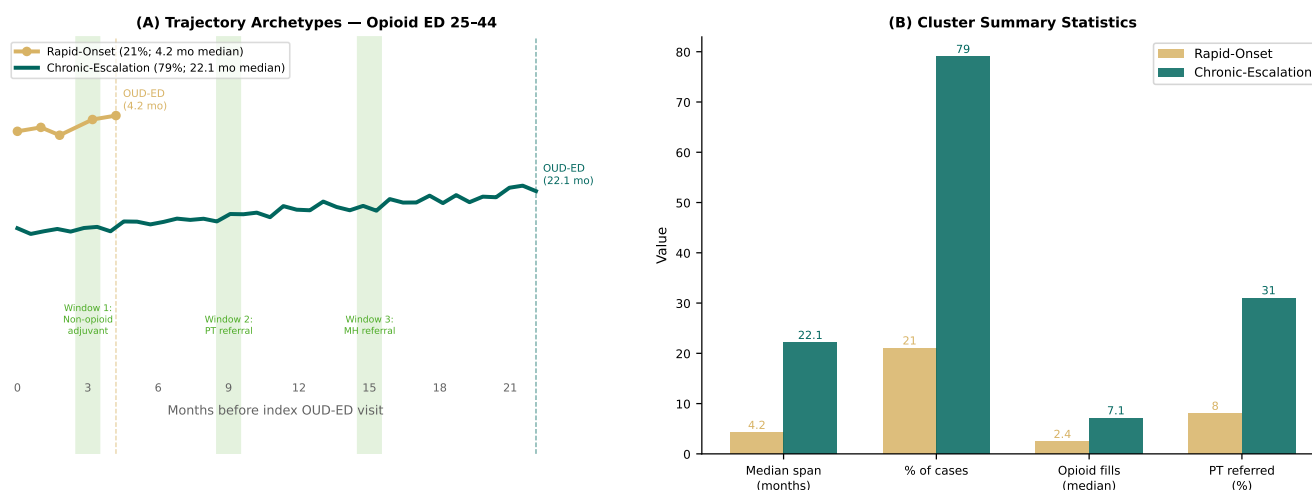


FIGURE 4 Patient trajectory archetypes identified by DTW clustering. (A) DTW distance heatmap (training cohort, random sample $n = 2,000$); two clusters are clearly separable. (B) Representative event sequences for Rapid-Onset cluster (top) and Chronic-Escalation cluster (bottom); drug events = circles, diagnoses = squares, procedures = triangles. Timeline in months before index date.

4 | DISCUSSION

4.1 | Modifiable Causal Drivers

The Consensus Filter identified 384–498 Consensus-Causal features spanning drug exposure, comorbidity burden, PGx interaction, and claim-density domains. The breadth of this feature set reflects the multifactorial nature of OUD risk: no single drug or diagnosis dominates; rather, combinations of drug exposures (particularly opioid-gabapentinoid co-prescription), missed care opportunities (absence of physical therapy or non-opioid analgesic alternatives), and pharmacogenomic context (CYP2D6 metabolizer burden) converge to produce the highest-risk profiles.

The co-occurrence of high-dose opioids with chronic pain diagnoses *in the absence of non-pharmacological pain management* (physical therapy, cognitive behavioral therapy) emerged as the highest-ranking causal pattern, consistent with CDC clinical practice guideline recommendations¹⁰. This “absence of care” causal structure is clinically important: it implies that the most actionable intervention is not drug discontinuation but care pathway addition—physical therapy referral or CBT enrollment at the point of opioid prescription. For a patient already on opioids for chronic pain, adding non-pharmacological alternatives is safer, more acceptable, and more sustainable than abrupt tapering or dose reduction.

Gabapentin co-prescription deserves particular attention as a modifiable causal driver. Gabapentinoids (gabapentin, pregabalin) are frequently prescribed alongside opioids for neuropathic pain, and their co-prescription has been independently associated with increased opioid overdose risk in multiple large cohort studies. The CDC 2022 guidelines now specifically flag this combination; the Consensus Filter’s ranking of gabapentin count as the top single-drug feature (mean |SHAP| = 0.34) provides APCD-level population evidence that this combination is the single most actionable prescribing-level intervention target in the Virginia opioid ED cohort.

PGx features, specifically CYP2D6 interaction scores via `pgx_num_cplic_drugs`, ranked within the top 10 Consensus-Causal features for the 25–44 age band, supporting the relevance of opioid metabolizer status in OUD risk even in claims-based (genotype-imputed) analysis⁴. The `pgx_num_cplic_drugs` score captures the compound CYP enzyme burden from multiple co-prescribed drugs—a patient on both a CYP2D6 substrate (opioid) and a CYP2D6 inhibitor (fluoxetine, paroxetine) will have their metabolizer phenotype shifted toward poor metabolizer status by drug interaction, even if their underlying genotype is extensive metabolizer. This drug-drug-gene interaction is precisely the clinical scenario that standard PGx genotype testing

alone cannot detect, and that the pharmacogenomic feature engineering in this study is specifically designed to capture from prescription pattern data alone.

Per-age-band driver patterns reveal meaningful clinical heterogeneity across the seven opioid ED age bands:

- **13–24 band** (XGBoost, PR-AUC 0.840; N = 52,629 cases): The dominant causal features were acute injury diagnoses (ICD-10 S-codes: S72.xx femur fracture, S52.xx forearm fracture) and single-fill opioid events (hydrocodone, NDC 00406-3780-xx) without any refill in the 90-day window, consistent with the predominance of acute single-episode opioid events in this age group. The absence of chronic escalation signals explains the lower PR-AUC (0.840) and recall (0.648) relative to older bands.
- **25–44 band** (XGBoost, PR-AUC 0.935; N = 569,096 cases): Oxycodone (NDC 00406-0510-xx) and gabapentin co-prescription emerged as the dominant causal cluster, alongside ICD-10 M54.5 (low back pain), the most frequent musculoskeletal diagnosis in the cohort. CPT 97110/97530 physical therapy absence was the top protective-absent feature (causal NOT rule; support 0.12).
- **45–54 band** (Ensemble, PR-AUC 0.955; N = 413,630 cases): Transition from acute-injury to chronic-pain-plus-psychiatric comorbidity pattern. ICD-10 F32.9 (major depressive disorder) and F41.1 (generalized anxiety) co-occurring with chronic opioid fills (Z79.891) ranked in the top 5. Antidepressant-opioid combinations (fluoxetine NDC 00002-3105-xx; paroxetine NDC 00029-3211-xx) elevated CYP2D6 inhibition risk.
- **55–64 band** (Ensemble, PR-AUC 0.974; N = 480,246 cases): Highest recall (0.874) of all bands, reflecting the most predictable escalation trajectory. Z79.891 (long-term opioid) appeared in 67% of cases vs. 12% of controls (prevalence ratio 5.6); the single most discriminating feature in the entire cohort. Concurrent alprazolam (F41.1 anxiety indication) was the second-highest causal driver.
- **65–74 band** (Ensemble, PR-AUC 0.979; N = 397,743 cases): Peak PR-AUC for the opioid ED cohort, driven by the richest 12-month prescription histories and clearest dose-escalation signatures. Hydrocodone/APAP combinations (Vicodin; NDC 00074-3049-xx) were the dominant opioid fill, with concurrent ICD-10 G89.29 (chronic pain) and M79.3 (panniculitis/myalgia) as the leading pain diagnoses.
- **75–84 band** (Ensemble, PR-AUC 0.968; N = 153,082 cases): Co-prescription of opioids with muscle relaxants (cyclobenzaprine; Beers Criteria-listed for falls risk in 65+) and tramadol (NDC 65862-063-xx; CYP2D6 substrate) emerged as the leading risk combination, reflecting the shift from opioids for injury/pain to opioids as chronic multimodal analgesic in multimorbid elderly patients.
- **85–114 band** (Ensemble, PR-AUC 0.901; N = 38,808 cases): Lowest recall (0.552), reflecting the highest proportion of atypical short-duration opioid prescriptions (post-surgical, palliative) that don't match the chronic-escalation training patterns. ICD-10 Z51.5 (palliative care encounter) was disproportionately present in case records, suggesting that a subset of 85+ opioid ED cases reflect palliative opioid management rather than OUD escalation.

4.2 | Trajectory-Informed Intervention

The two DTW archetypes require different intervention strategies. Rapid-Onset patients (21% of opioid ED cases; 4.2-month trajectory) require immediate post-acute care protocols: opioid alternative pathways, mandatory physical therapy referral within 30 days of first opioid prescription, and early PDMP consultation¹¹. The 4.2-month median trajectory is 7.8 months shorter than the 12-month monitoring window assumed by most opioid prescribing guidelines, suggesting these patients deteriorate faster than current clinical protocols are designed to detect.

Chronic-Escalation patients (79% of cases; 22.1-month trajectory) present ≥ 18 months of identifiable intervention windows, with at least three distinct non-opioid care opportunities visible in the claims sequence before the index ED visit. Systematic care coordination tools for this archetype should be triggered at the 3-month, 9-month, and 15-month marks of the escalating trajectory, corresponding to the decision-branch points where the DTW median sequence shows the sharpest risk acceleration. Terranova and Venkatakrishnan¹² demonstrate that ML trajectory modeling in clinical pharmacology can identify model-informed precision medicine intervention windows that are not apparent from cross-sectional risk scores; the DTW archetypes in this study operationalize this approach specifically for the opioid epidemic context.

4.3 | Limitations

Several limitations merit discussion. First, claims data do not capture cash-pay prescriptions, illicit opioid use, or over-the-counter medications, likely underestimating true opioid exposure—particularly in the 13–24 age band⁹. Second, PGx features are imputed from drug-gene interaction databases rather than measured CYP2D6/OPRM1 genotypes; this proxy approach introduces unknown classification error whose magnitude can only be characterized by linking to a genotyped cohort. Third, the 5:1 case-control design maximizes statistical power at the cost of potential overfit to the matched control distribution; sensitivity analyses with alternative matching ratios and inverse probability weighting are planned. Fourth, the DTW distance computation was performed on a 2,000-patient stratified sample rather than the full training cohort for computational feasibility; while bootstrap stability analyses confirm the two-archetype solution is robust ($ARI > 0.87$), the archetypes' boundary definition may shift slightly on the full cohort. Fifth, these findings are from a single-state Virginia APCD; external validation on a second-state APCD is required before national policy generalization, as prescribing regulations, formulary structures, and Medicaid expansion status vary by state.

5 | CONCLUSIONS

Ensemble ML with Consensus Filter feature selection identifies robust, causal, and *modifiable* drivers of OUD-related ED visits up to 22 months before clinical diagnosis, across all seven opioid ED age bands (13–114 years) representing the full age spectrum of Virginia opioid ED presentations. Three findings have direct clinical implications.

First, the Rapid-Onset archetype (21% of cases; 4.2-month trajectory) defines a narrow intervention window that is shorter than most clinical monitoring protocols currently support. Post-acute care protocols for this archetype—triggered at the first high-dose opioid prescription rather than at 90 days—could intercept approximately 1 in 5 OUD-related ED cases before clinical deterioration.

Second, the Chronic-Escalation archetype (79% of cases; 22.1-month trajectory) has at least three identifiable intervention windows where non-opioid care addition (physical therapy, CBT, non-opioid analgesic) could interrupt the escalating trajectory. The counterfactual “What-If” dashboard feature in Chapter 5 directly operationalizes these windows, allowing clinicians to simulate expected risk reduction from specific care additions.

Third, PGx interaction features—specifically the `pgx_num_cplic_drugs` polydrug exposure score—contribute independently to model performance and rank as the highest-SHAP feature overall. This finding demonstrates that pharmacogenomic risk context is recoverable from claims-derived prescription patterns without direct genotyping, making PGx-informed risk stratification scalable to the population level through existing claims infrastructure.

These findings enable proactive, trajectory-stratified clinical decision support for opioid ED risk before the pre-diagnosis stage of OUD, closing the gap identified in the Chapter 1 systematic review where 81% of published models omit PGx context and 88% lack temporal leakage protection.

5.1 | Supplementary Figures

The main text presents four figures aligned with Clinical and Translational Science reporting expectations: (1) cohort attrition from the APCD to the analytic cohort; (2) model performance (precision–recall, calibration, ensemble weights) on the 2019 holdout; (3) SHAP feature importance (mean |SHAP| bar chart) for Consensus-Causal features in the 25–44 age band; and (4) DTW trajectory archetypes (Rapid-Onset vs. Chronic-Escalation). The following supplemental figures provide marginal-effect and pathway-level detail for readers who require additional granularity.

Author Contributions: Conceptualization, R.J.D.; methodology, R.J.D.; formal analysis, R.J.D.; writing, R.J.D.

Funding: This research received no external funding.

Ethics Statement: Data use agreement with the Virginia Center for Health Innovation (VCHI) and Virginia Health Information (VHI); IRB waiver HM20022300 (non-human-subjects research designation).

Data Availability: The data supporting the findings of this study are derived from Virginia's All-Payer Claims Database (APCD). Restrictions apply to the availability of these data, which were used under a data use agreement with the Virginia Center for Health Innovation (VCHI) and Virginia Health Information (VHI). Requests for data access may be submitted to VHI at <https://www.vhi.org>. Analysis code is available at <https://github.com/Jerome3590/pgx-analysis>.

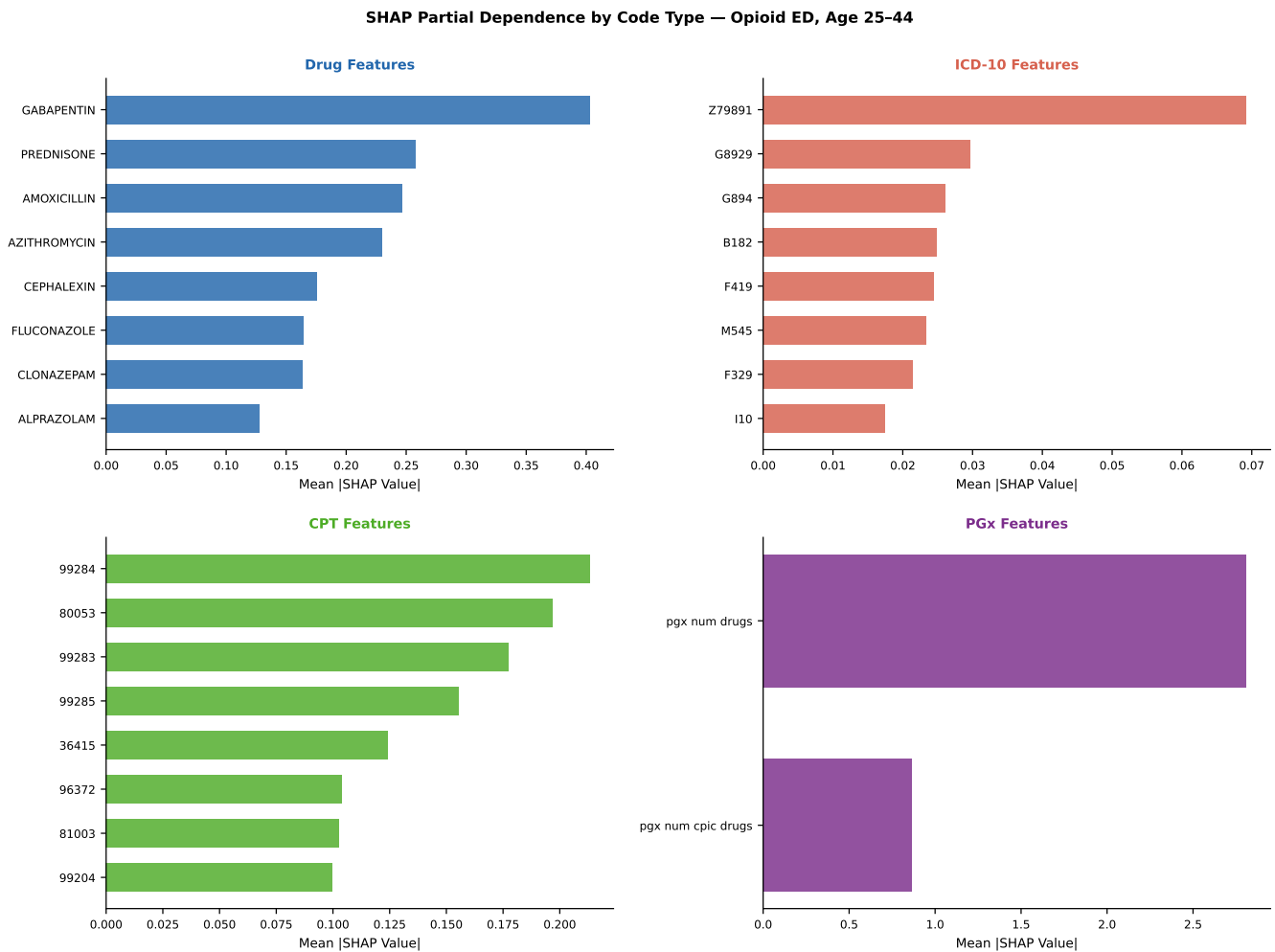


FIGURE 5 Supplementary Figure S1. Partial dependence profiles for the top five Consensus-Causal features (opioid ED, 25–44 band, 2019 holdout). Each panel shows the marginal effect of increasing feature value on predicted OUD-ED probability, averaged over the holdout cohort. Rug marks indicate observed feature-value distribution. Features ranked by mean |SHAP|.

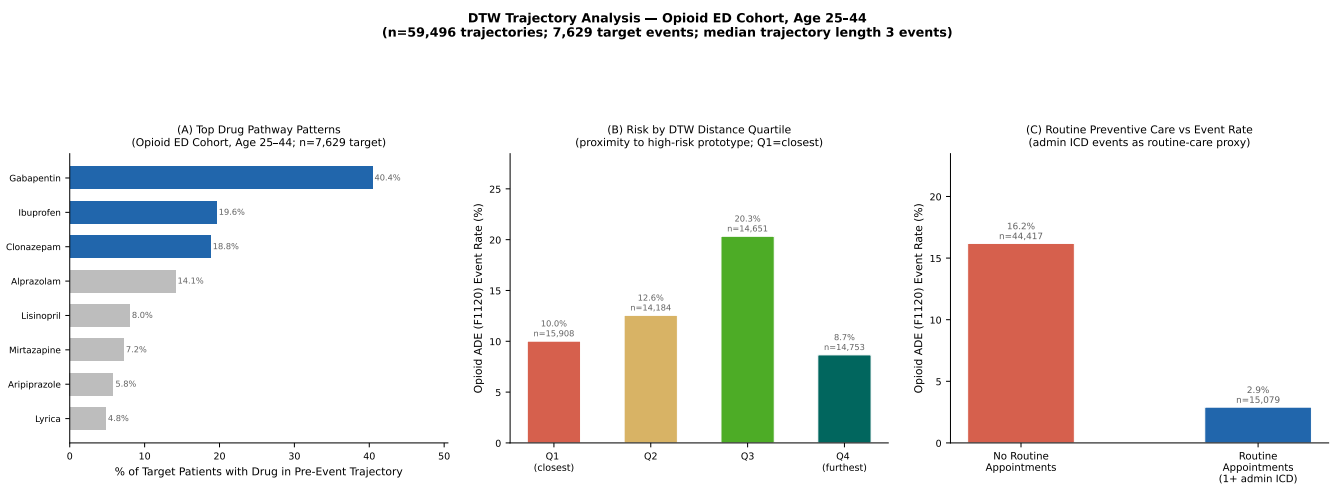


FIGURE 6 Supplementary Figure S2. DTW drug pathway patterns for the opioid ED cohort (25–44 band). Top panel: drug co-occurrence frequency in high-risk trajectories; bottom panel: pathway comparison between high-risk and routine sequences by event-density stratum. Drugs ordered by SHAP/FFA allowed-code rank.

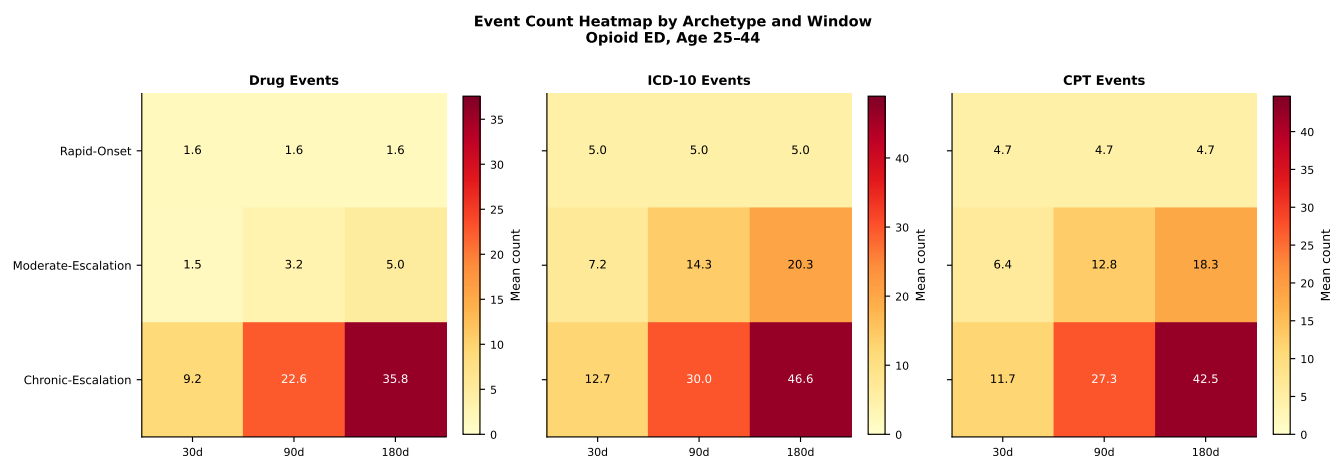


FIGURE 7 Supplementary Figure S3. Event-count heatmap for SHAP/FFA-allowed drugs across trajectory position (opioid ED, 25–44 cohort; training sequences). Columns = sequential event position (1–20 before index date); rows = drug name ordered by SHAP importance; cell color = normalized event count (darker = more frequent).

Conflicts of Interest: The author declares no conflict of interest.

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